FINAL-YEAR COMPUTER ENGINEERING · SIIT, THAMMASAT UNIVERSITY

Singha Junchan

สิงหา จุลจันทร์ / Pathum Thani, Thailand

I build access control for cloud and AI systems.

Scan or click for the full portfolio, code links and contact.

singhaju.github.io

linkedin.com/in/singha-junchan

github.com/singhaju

Open to: security internships from January 2027, in Thailand or abroad full-time roles from May 2027

- **1st in Thailand**: national champion, Thailand Olympiad in Informatics (TOI18), 2022.
- **Best Paper Award**, IEEE JCSSE 2026, for **U-HAP**, Kubernetes authorization (second author).
- **9 security research projects** on **Kubernetes**, **zero trust**, **zero-knowledge proofs**, **post-quantum** cryptography and **AI agents**; first author of **FEDAC** (under review, IEEE TDSC).
- **Security intern, Allianz Ayudhya**, mentored by the CISO: built **two AI agents for third-party risk**.
- **Team lead, TU Alert**: campus incident reporting in **LINE**, built for Thammasat University (launching soon).
- Final-year **Computer Engineering**, SIIT, Thammasat University: **GPA 3.67**, **full scholarship**, graduating **May 2027**.

COMPETITIVE PROGRAMMING
#1 in Thailand
TOI18 national champion, 2022, after 1,000+ hours of training

CLIENT PROJECT
TU Alert
built for **Thammasat University**; **I lead the team**; security built in; launching soon

RESEARCH
9 research projects
2 accepted (one **Best Paper**), 3 submitted, 4 in preparation; first author on 8

INDUSTRY
2 internships
Allianz Ayudhya (security, mentored by the CISO) and **Amity Solutions** (AI chatbots)

LEADERSHIP AND TEACHING
11 student events
led SIIT First Meet 2024 (about 500 people) and organised 10 more; founded **two coaching platforms**

CERTIFICATION
ISC2 Certified
Certified in Cybersecurity (CC), 2025; CompTIA Security+ planned Dec 2026

01 What I bring

Security engineering, end to end
I **design security mechanisms and build them**: a zero-trust framework with a zero-knowledge verifier (**first author**) and a Kubernetes authorization webhook (**Best Paper**, second author).

Zero trust · Applied cryptography · Kubernetes · Go and Rust

Identity, access and governance
Who may do what, checked on every request and **provable later**. Growing into **governance**: policy, third-party risk and audit.

IAM · RBAC / ABAC · Audit trails · Third-party risk

Cloud-native by default
I test ideas on **real cloud deployments**: Kubernetes on **AWS EKS**, prototypes on **EC2**, containers with **Docker**.

Kubernetes · AWS EKS and EC2 · Docker · Envoy and Istio

AI agents, and how to limit them
I **build AI agents** that do useful work, and **research controls** that keep them inside what a human allowed, even under attack.

LLM agents · MCP · Prompt injection · Agent permissions

Leads, teaches and explains
I **start things and run teams**: TU Alert, 11 student events, two coaching platforms. I explain hard ideas simply.

Team lead · C++ teaching · Event organising

02 Experience

Team Lead and Developer, TU Alert

● Launching soon

Thammasat University (client project) / Rangsit campus, Pathum Thani — 2026

- Thammasat University hired our student team to build **TU Alert**, a **LINE-based campus incident-reporting system**. **I lead the team** and write code.
- **TypeScript** monorepo (Hono, Bun, Prisma, MariaDB, Redis, Next.js, LINE LIFF). **Security built in**: four roles, second-step staff sign-in, data retention and erasure, **Semgrep and gitleaks** in CI.

KNOWLEDGE: Secure web development · Role-based access control · Data retention and erasure · Team leadership

TECH: TypeScript · Next.js · Hono · Bun · Prisma · MariaDB · Redis · LINE LIFF · Semgrep · gitleaks

Information Security Intern

Allianz Ayudhya / Bangkok, Thailand — Jun to Jul 2026

- Mentored by the **CISO**, I built **two AI agents for third-party risk**: a **questionnaire agent** that answers due-diligence questionnaires with evidence, a reason and a **confidence score** per answer, and a **guidance agent** that walks business owners through each step.
- **Tested on real, anonymized cases** with the Information Security team; demonstrated the questionnaire agent (6 Jul) and **presented both** (31 Jul 2026).

KNOWLEDGE: Third-party risk management · Security governance · Due-diligence questionnaires · Gap and fit-gap analysis · AI agents

Founder, Curriculum Author and Lead Instructor

TOI Tutor and ABIG Center (online) / toitutor.in.th — 2022 to present

- **Founded two online platforms** that coach students for Thailand's national programming olympiads; **coached hundreds of students** since 2022.
- Wrote and taught **COM300**, a full C++ and algorithms course: about 270 pages and **128 recorded lessons**.

KNOWLEDGE: Algorithms and data structures · Competitive programming · Curriculum design · Teaching

TECH: C++

R&D Intern, AI research team

Amity Solutions / Bangkok, Thailand — Jul to Aug 2023

- On the AI research team, I **built chatbots end to end** with **large language models (LLMs)** and a **knowledge base**, including the **AI pipeline** behind them; also data analysis and prompt engineering.

KNOWLEDGE: Large language models · Chatbots · Knowledge bases · AI pipelines · Prompt engineering · Data analysis

03 Research

Research supervisor: Prof. Somchart Fugkeaw, School of ICT, SIIT.

9 research projects: 2 accepted, 3 submitted, 4 in preparation. First author on 8. One question links most of them: can a security check be cheap enough to run on every request, and still be provable afterwards?

FEDAC

● Under review · IEEE TDSC

ZERO-TRUST ACCESS CONTROL — First author of 4

Verify the caller once, then check cheaply on every call. A hub picks the sign-in strength per service, up to a **zero-knowledge proof**, and issues a short-lived token; an **Envoy sidecar** checks it locally. Policies on **IPFS** are anchored on a **blockchain**, so every change can be audited.

My part: Research design and the full implementation, as first author of 4.

KNOWLEDGE: Zero trust · Zero-knowledge proofs · Service mesh · Blockchain audit · Cloud benchmarking

TECH: Go · Rust · Groth16 zk-SNARK · Envoy + Istio · IPFS · AWS EKS

RESULT **1.4 ms** median authorization latency on AWS EKS, about 7,100 requests per second (paper under review)

U-HAP

● Presented · IEEE JCSSE 2026 · Best Paper Award

ZERO-TRUST ACCESS CONTROL — Second author of 4 · presented the paper

Authorization as a compiler problem. One graph model covers **RBAC, ABAC, ACL and deny rules**; policies are compiled ahead of time into per-resource indexes, so a request checks only the rules that can apply. **Deny always wins**, so every decision is deterministic.

My part: Co-developed the webhook and policy model, checked it against OPA, and presented the paper.

KNOWLEDGE: Access control (RBAC, ABAC, ACL) · Kubernetes admission · Policy compilation · Graph models

TECH: Python · Kubernetes webhook · OPA (baseline)

RESULT **~2.7× faster** policy updates reach a first decision vs OPA at 2,000 policies, with identical decisions

ZKAudit
● Accepted · KST 2027

ZERO-KNOWLEDGE AUDIT — First author

Prove you may audit, without revealing who you are. The auditor proves its right to read cloud logs in **zero knowledge**; proof checks are shared across verifiers, and the logs are checked against **Merkle commitments**, so tampering shows.

KNOWLEDGE: Zero-knowledge proofs · Cloud audit logs · Merkle commitments · Privacy

TECH: Rust · Groth16 zk-SNARKs · AWS EC2

Post-quantum authorization across Kubernetes clusters
● Submitted

POST-QUANTUM SECURITY — First author

Every decision carries a post-quantum proof of its scope, signed with **ML-DSA**, keys agreed with **ML-KEM**. State is anchored on a **ledger** every epoch, so a failure or rollback cannot quietly widen permissions.

KNOWLEDGE: Post-quantum cryptography · Multi-cluster Kubernetes · Authorization · Distributed ledgers

TECH: Go · ML-DSA · ML-KEM · Hyperledger Fabric · AWS EKS

Self-healing fog computing for industrial IoT
● Submitted

RESILIENT CLOUD AND EDGE — First author

Edge servers that share work like a swarm, with no central controller. Nodes swap small hints, pull work when they have spare capacity and **recover after failures**; shared results are protected with **homomorphic encryption** and **Merkle proofs**.

KNOWLEDGE: Fog and edge computing · Industrial IoT · Fault tolerance · Homomorphic encryption

TECH: Python · gRPC · Paillier · Merkle proofs · AWS EC2

MoCA: AI containment
● In preparation

AI-AGENT SECURITY — First author

Check an AI agent's authority when it acts, not when it plans. Permissions **only shrink** as agents delegate, revocation can be verified, and text from an untrusted page (a **prompt injection**) cannot push an action past its limit.

KNOWLEDGE: AI-agent security · Delegation and revocation · Prompt injection

TECH: Python · MCP tool gateway · Attack test harness

ZK-ALBA
● In preparation · journal paper

ZERO-KNOWLEDGE AUDIT — First author

Audit proofs bound to what each auditor may see. Each zero-knowledge proof is tied to the auditor's scope, and **load-aware verifier shards** reject reused proofs.

KNOWLEDGE: Zero-knowledge proofs · Scoped audit · Load balancing · Replay protection

TECH: Rust · Sharded verifiers · AWS EC2

Threshold-gated encrypted SIEM
● In preparation

SECURITY OPERATIONS — First author

Catch an attack across companies without sharing readable logs. Companies correlate **encrypted** security logs, and tracing across them opens only when **several companies independently** report the same signal.

KNOWLEDGE: SIEM and threat detection · Applied cryptography · Threshold schemes · Privacy across companies

TECH: Python · Detection pipeline · LANL enterprise dataset

Fresh answers from elastic multi-cloud services
● In preparation

RESILIENT CLOUD AND EDGE — First author

Scale stateful services up and down without serving stale data. As replicas are added or removed across clouds, every answer carries a small **freshness proof**.

KNOWLEDGE: Multi-cloud systems · Elastic scaling · Freshness proofs

TECH: Python · Sparse Merkle trees · Multi-cloud deployment

How I work on each project

1. Find the gap in published designs
2. Design the scheme
3. Build the prototype
4. Measure it on AWS or real data
5. Write the paper

04 Hands-on security

NCSA CTF Thailand Tournament 2026	Thailand Cyber Top Talent 2026, Aug 2026. Competed in a team on web, cryptography, digital forensics, OSINT and mobile challenges.
SOSECURE Road to Cybersecurity 2025 (GEN6)	Eight modules with labs: routing and firewalls, Linux hardening, Python security tools, Metasploit and web attacks, SOC log analysis in Splunk and Wazuh.
Ongoing practice	TryHackMe and Hack The Box. Nmap scanning in a Kali Linux home lab. DataCamp courses on Azure and AWS (2025).

05 Education and leadership

Education

B.Eng. in Computer Engineering — Aug 2023 to May 2027 (expected)
Minor in Cloud Computing and Cybersecurity. Sirindhorn International Institute of Technology (SIIT), Thammasat University GPA 3.67 / 4.00, first-class honours expected.

Leadership

11 student events organised · ~500 participants at SIIT First Meet 2024 · 100 staff at SIIT First Meet 2024

President, Sub Student Committee 32, SIIT — 2024 to present
Leads a committee that represents the student body and organises academic and social events.

Lead organiser, SIIT First Meet 2024 — 2024
About 500 participants and 100 staff. Organiser of 10 more student events, 2023 to 2025.

Project manager and product owner, Novel Nest — Team project
Led a team of four to build a digital library and creative hub for readers and writers.

06 Awards and certifications

2022 — **National Champion: 1st place in Thailand and Gold Medal**
Thailand Olympiad in Informatics (TOI18)

2025 — **1st place overall**
Image Recognition Programming Challenge, Osaka Institute of Technology and SIIT

2022 — **Bronze Medal and Popular Vote**
SCIUS Forum 2022, Automated Cat Feeder

2026 — **Best Paper Award**
IEEE JCSSE 2026, Bangkok, for U-HAP (second author of 4)

2023 to 2027 — **Full scholarship**
4-Year Continuing Academic Outstanding Student Program, SIIT

CERTIFICATIONS ✓ **ISC2 Certified in Cybersecurity (CC)** 2025 planned **CompTIA Security+** Exam planned Dec 2026

07 Projects

TU Alert, Novel Nest, Arduino retro console, Automated Cat Feeder, Velour Mall.

08 Skills

Used in my research	Zero Trust · Kubernetes (RBAC, ABAC, webhooks) · AWS (EKS, EC2) · Docker · Envoy and Istio · OPA / Rego · IPFS · Hyperledger Fabric
AI agents and AI security	LLM agent pipelines · MCP gateway · Prompt-injection testing · Prompt engineering · NLP chatbots
Shipping and operations	GitHub Actions CI · Semgrep · gitleaks · Playwright · pm2 on Linux · Backups and restore
Cryptography	zk-SNARKs (Groth16) · Merkle commitments · ML-DSA · ML-KEM
Security tools (labs and CTFs)	Nmap · Wireshark · Burp Suite · Metasploit · OSINT · OWASP Top 10
Programming	Python · Go · Rust · C++ · C · Java · Bash · TypeScript · SQL
Also familiar with	Splunk and Wazuh (SOC labs) · Azure and AWS basics (DataCamp)
Languages	Thai (native) · English (fluent) · Chinese (basic)

09 References

People I have worked with, who agreed to be named. Contact details on request.

Apichai Wongchindakhun
Chief Information Security Officer (CISO)
Allianz Ayudhya
Mentored me during my information security internship (Jun to Jul 2026), where I built and presented two AI-agent prototypes for third-party risk.
Internship mentor

Prof. Somchart Fugkeaw
Professor, School of ICT
SIIT, Thammasat University
Supervises my research and co-authors my papers, including FEDAC (under review, IEEE TDSC) and ZKAudit (accepted, KST 2027).
Research supervisor

Pachara
Co-developer, TU Alert
SIIT, Thammasat University
Builds TU Alert with me for Thammasat University.
Project partner

Let's talk

Looking for an internship from January 2027 and full-time roles from May 2027: security engineering, cloud security or AI security.

singhaju.github.io
github.com/singhaju
linkedin.com/in/singha-junchan